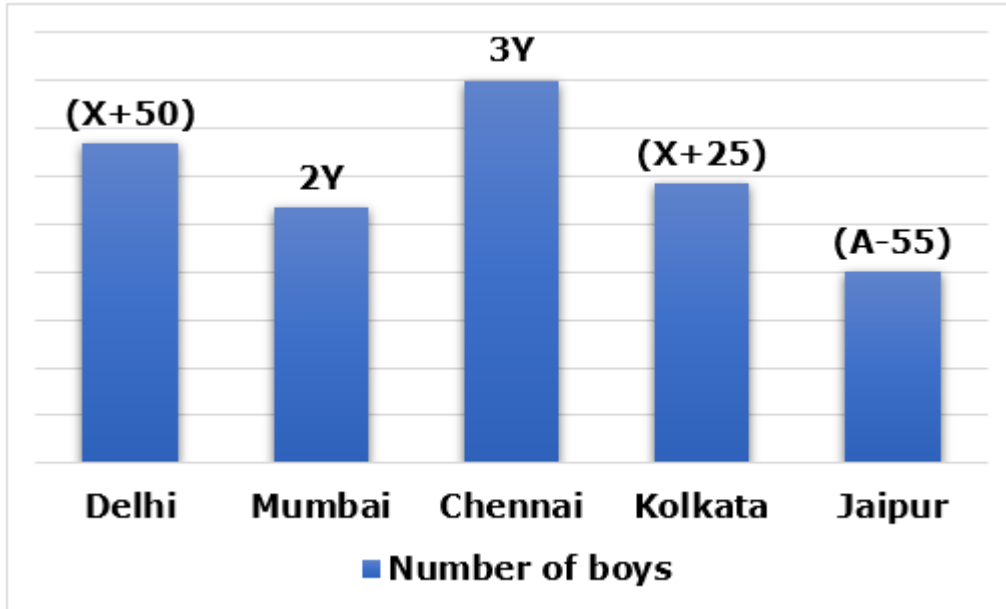


1. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the number of boys in the school in five different cities Delhi, Mumbai, Chennai, Kolkata and Jaipur.



Notes:

The total number of students in the school = the number of boys in the school + the number of girls in the school

The total number of students (boys + girls) in the school in Delhi, Mumbai, Chennai, Kolkata and Jaipur 320, 240, 400, 300 and 180 respectively.

The number of girls in the school in Delhi is 50% more than Mumbai.

The ratio of the number of girls in the school in Chennai and Kolkata is 32:25.

The number of girls in Jaipur is $(B+5)$.

The difference between A and B is 120. ($A > B$)

If 10% of the total number of boys in the school in Delhi did not appear in the test and the ratio of the number of boys who passed and failed in the test is 2:1, then find the number of boys who failed in the test is what percent of the number of girls in the school in Delhi?

- 25%
- 50%
- 75%
- 30%
- None of these

2. Questions

If the number of girls who newly joined the school in Mumbai is half of the number of girls who left the school in Chennai, then the ratio of the number of girls in the school in Mumbai and Chennai will be 5:6. Find the number of girls who left the school in Chennai.

- a. 32
- b. 24
- c. 56
- d. 40
- e. 20

3. Questions

In another city Pune, if the total number of students in the school is $[A+25]$ and the average age of all boys and girls is 12 years and 10 years respectively, then the total age of all students (boys + girls) will be 2300 years. Find the difference between the number of boys and girls in the school in Pune.

- a. 100
- b. 112
- c. 150
- d. 144
- e. None of these

4. Questions

In Kolkata, if the number of boys who speaks both English and Hindi is 63 and the ratio of the number of boys who speaks only English and only Hindi is 11:5, then find the total number of boys who speaks English in the school in Kolkata.

Note: All boys either speak English or Hindi.

- a. 98
- b. 105
- c. 140
- d. 77
- e. 120

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5. Questions

In Jaipur, if the total number of teachers is $\frac{1}{3}^{\text{rd}}$ of the total number of students, and the ratio of the number of male and female teachers in the school is same as the ratio of the number of boys and girls in the school, then find the number of female teachers in the school in Jaipur.

- a. 40
- b. 50
- c. 10
- d. 30
- e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

Three persons P, Q and R started a business with certain investment in two different years 2015 and 2016.

2015: The ratio of the initial investment made by P, Q and R is 3:2:4. After 5 months, P withdrew Rs. 200, Q added Rs. 600 and R withdrew $\frac{1}{4}^{\text{th}}$ of his initial investment. After another 4 months, P increased his investment by 50%, Q withdrew Rs. 300 and R added Rs. 800. At the end of the year, the profit ratio of Q and R is 129: 188, respectively.

2016: The initial investment made by P, Q and R is Rs. 900, Rs. 700 and Rs. 1200 respectively. After few months, P withdrew $\frac{1}{3}^{\text{rd}}$ of his initial investment, Q added Rs. 300 and R withdrew 25% of his initial investment. At the end of the year, P and Q received equal amount of profit.

Note:

The profit will be divided in the ratio of the total investment made by each person.

In 2016, if 10% of the total profit received by all three persons together is donated to the charity, then difference between the profit share received by R and the amount of profit donated to the charity will be Rs. 64000. Find the total profit earned by all three persons together in 2016.

- a. Rs. 200000
- b. Rs. 180000
- c. Rs. 240000
- d. Rs. 120000
- e. Rs. 300000

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7. Questions

Which of the following is/are correct?

I: The initial investment made by P in 2015 is Rs. 1200.

II: The difference between the initial investment made by Q and R in 2015 is Rs. 800.

- a. Only I
- b. Only II
- c. Either I or II

- d. Both I and II
- e. Neither I nor II

8. Questions

In 2016, if another person S also invested Rs. $[4Y]$, but after 9 months, he withdrew $\frac{1}{4}^{\text{th}}$ of his initial investment, then the ratio of the profit share of Q and S at the end of the year will be 16:15. Find the value of $[20\% \text{ of } 2Y]$.

- a. 66
- b. 54
- c. 80
- d. 92
- e. 40

9. Questions

In 2017, if P started the business with the investment of Rs. 500, but after $[Z]$ months, Q also joined him with Rs. 1000 and P kept his initial investment unchanged, then the ratio of the profit received by P and Q at the end of the year will be 6:7. After how many months did Q joined the business.

- a. 8 months
- b. 5 months
- c. 7 months
- d. 6 months
- e. None of these

10. Questions

In 2015, if the difference between the profit received by P and R is Rs. 43000, then find the profit received by Q in the same year.

- a. Rs. 129000
- b. Rs. 114000
- c. Rs. 150000
- d. Cannot be determined
- e. Rs. 145000

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11. Questions

Read the following information carefully and answer the questions based on it.

The funnel chart given below shows the total number of voters registered in two different villages A and B. It also shows the percentage of the number of voters who casted their votes out of the total number of voters registered, the number of valid votes casted by the voters and the ratio of the number of valid votes casted by males and females in each village.

	Village A	Village B
Total number of voters registered	2000	100Q
Percentage of the number of voters casted their votes	2P%	90%
Number of valid votes casted by the voters	1400	960
Ratio of the number of valid votes casted by males and females	M:3	N:3

Notes:

- i). The total number of voters registered = the number of voters who casted their votes + the number of voters who did not cast their votes
- ii). The total number of voters casted their votes = the number of valid votes casted by the voters + the number of invalid votes casted by the voters
- iii). The total number of valid votes casted by the voters = the number valid votes casted by males + the number of valid votes casted by females
- iv). In village A, the number of invalid votes casted by the voters is half of the number of voters who did not cast their votes.
- v). In village B, the number of voters who did not cast their votes is equal to the number of invalid votes casted by the voters.
- vi). The number of valid votes casted by females in village A is equal to the number of valid votes casted by males in village B.
- vii). The number of valid votes casted by females in village B is 55% less than the number of valid votes casted by males in village A.

In village A, if the ratio of the number of males and females who casted their votes (valid + invalid) is 19:13, then find the difference between the number of invalid votes casted by males and females in village A.

- a. 120
- b. 60
- c. 75

d. 100

e. 50

12. Questions

In village B, if the number of valid votes casted to party X is 5 times the number of ivalid votes casted to the same party and the number of valid votes casted to party Y is 440 more than the number of invalid votes casted to party Y, then find the ratio of the total number of votes casted (valid + invalid) to party X and Y in village B.

a. 5:4

b. 3:2

c. 2:1

d. Cannot be determined

e. None of these

13. Questions

In another village C, if the total number of voters who casted their votes is [150N], out of which $\frac{1}{5}$ th is invalid, which is 100 less than the number of voters who did not cast their votes, then find the total number of voters registered in village C is what percent of village B?

a. 62.50%

b. 83.33%

c. 75%

d. 50%

e. 77.77%

14. Questions

Which of the following is/are definitely true?

I: $[P/N] = 8$

II: $[M+5] = Q$

III: $[P-M] = 6^2$

a. Only III

b. Both I and II

c. Both I and III

d. All I, II and III

e. None is true

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15. Questions

In village A, if the number of valid votes casted to losing candidate is 12.50% of the total number of voters who casted their votes, then find the number of votes by which winning candidate won the election in village A?

- a. 800
- b. 1200
- c. 600
- d. 1000
- e. None of these

16. Questions

Two quadratic equations are given below.

Equation I: $A^3 - 28A^2 + 187A = 0$

Equation II: $B^3 - 32B^2 + 192B = 0$

- a. $A > B$
- b. $A < B$
- c. $A \geq B$
- d. $A \leq B$
- e. $A = B$, or relation cannot be determined

17. Questions

Two quadratic equations are given below.

Equation I: $P^2 + MP - 108 = 0$

Equation II: $Q^2 - NQ - 240 = 0$

$(M-1)^3 = N$

Both M and N are single-digit natural numbers.

M is odd prime number.

Find the difference between the smaller roots of both equations.

- a. 12
- b. 0
- c. 4
- d. 18

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e. None of these

18. Questions

A quadratic equation is given below.

Equation: $K^2 - 17K + 70 = 0$

If there is another equation whose greater root is equal to the smaller root of the equation given above, then find the correct equation.

- a. $L^2 - 11L + 28 = 0$
- b. $L^2 - 18L + 77 = 0$
- c. $L^2 - 27L + 140 = 0$
- d. All of the above
- e. None of these

19. Questions

A quadratic equation is given below.

Equation: $X^2 - (A)X + 400 = 0$

The smaller root of the equation is (A-40).

Find the greater root of the equation given above.

- a. 20
- b. 50
- c. 80
- d. 40
- e. 10

20. Questions

A quadratic equation is given below.

Equation: $Y^2 - MY + N = 0$

The sum of the digits of greater root is equal to the smaller root, which is 5. The greater root of the equation is more than 20 but less than 30.

Find the value of $[N/(M-5)]$.

- a. 5
- b. 10

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- c. 23
- d. 32
- e. None of these

Explanations:

1. Questions

Delhi and Mumbai

The total number of students in the school in Delhi = 320

The number of boys in the school in Delhi = $(X+50)$

So, the number of girls in the school in Delhi = $\{320-(X+50)\} = (270-X)$

The total number of students in the school in Mumbai = 240

The number of boys in the school in Mumbai = $2Y$

So, the number of girls in the school in Mumbai = $(240-2Y)$

Chennai and Kolkata

The total number of students in the school in Chennai = 400

The number of boys in the school in Chennai = $3Y$

So, the number of girls in the school in Chennai = $(400-3Y)$

The total number of students in the school in Kolkata = 300

The number of boys in the school in Kolkata = $(X+25)$

So, the number of girls in the school in Kolkata = $\{300-(X+25)\} = (275-X)$

Now,

$$(270-X)/(240-2Y) = 150/100$$

$$(270-X)/(240-2Y) = 3/2$$

$$540 - 2X = 720 - 6Y$$

$$6Y - 2X = 180$$

$$3Y - X = 90 \dots\dots\dots(I)$$

And,

$$(400-3Y)/(275-X) = 32/25$$

$$10000 - 75Y = 8800 - 32X$$

$$75Y - 32X = 1200 \dots\dots\dots(II)$$

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By solving both equations (I) and (II), we get,

$$225Y - 75X = 6750$$

$$225Y - 96X = 3600$$

$$21X = 3150$$

$$X = 150$$

Now, put the value of X in equation (I), we get,

$$3Y - X = 90$$

$$3Y - 150 = 90$$

$$3Y = 240$$

$$Y = 80$$

Also,

The total number of students in the school in Jaipur = 180

The number of boys in the school in Jaipur = (A-55)

So, the number of girls in the school in Jaipur = (B+5)

Now,

$$(A-55) + (B+5) = 180$$

$$A + B = 230 \dots\dots\dots(III)$$

And,

$$A - B = 120 \dots\dots\dots(IV)$$

By solving both equations (III) and (IV), we get,

$$2B = 110$$

$$B = 55$$

Now, put the value of B in equation (III), we get,

$$A + B = 230$$

$$A + 55 = 230$$

$$A = 175$$

Delhi

The total number of students = 320

The number of boys = (X+50) = (150+50) = 200

The number of girls = (320-200) = 120

Mumbai

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The total number of students = 240

The number of boys = $2Y = (2 \times 80) = 160$

The number of girls = $(240 - 160) = 80$

Chennai

The total number of students = 400

The number of boys = $3Y = (3 \times 80) = 240$

The number of girls = $(400 - 240) = 160$

Kolkata

The total number of students = 300

The number of boys = $(X + 25) = (150 + 25) = 175$

The number of girls = $(300 - 175) = 125$

Jaipur

The total number of students = 180

The number of boys = $(A - 55) = (175 - 55) = 120$

The number of girls = $(180 - 120) = 60$

City	The total number of students (boys + girls)	The number of boys	The number of girls
Delhi	320	200	120
Mumbai	240	160	80
Chennai	400	240	160
Kolkata	300	175	125
Jaipur	180	120	60
Total	1440	895	545

Answer: B

Now, according to the question,

Delhi

The total number of boys = 200

The number of boys who appeared in the test = 90% of 200 = 180

So, the number of boys who failed in the test = $\frac{1}{3}$ of 180 = 60

The number of girls = 120

Therefore, the required percentage = $\left\{ \left(\frac{60}{120} \right) \times 100 \right\} \% = 50\%$.

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Hence, the correct answer is option B.

2. Questions

Delhi and Mumbai

The total number of students in the school in Delhi = 320

The number of boys in the school in Delhi = $(X+50)$

So, the number of girls in the school in Delhi = $\{320-(X+50)\} = (270-X)$

The total number of students in the school in Mumbai = 240

The number of boys in the school in Mumbai = $2Y$

So, the number of girls in the school in Mumbai = $(240-2Y)$

Chennai and Kolkata

The total number of students in the school in Chennai = 400

The number of boys in the school in Chennai = $3Y$

So, the number of girls in the school in Chennai = $(400-3Y)$

The total number of students in the school in Kolkata = 300

The number of boys in the school in Kolkata = $(X+25)$

So, the number of girls in the school in Kolkata = $\{300-(X+25)\} = (275-X)$

Now,

$$(270-X)/(240-2Y) = 150/100$$

$$(270-X)/(240-2Y) = 3/2$$

$$540 - 2X = 720 - 6Y$$

$$6Y - 2X = 180$$

$$3Y - X = 90 \dots\dots\dots(I)$$

And,

$$(400-3Y)/(275-X) = 32/25$$

$$10000 - 75Y = 8800 - 32X$$

$$75Y - 32X = 1200 \dots\dots\dots(II)$$

By solving both equations (I) and (II), we get,

$$225Y - 75X = 6750$$

$$225Y - 96X = 3600$$

$$21X = 3150$$

$$X = 150$$

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Now, put the value of X in equation (I), we get,

$$3Y - X = 90$$

$$3Y - 150 = 90$$

$$3Y = 240$$

$$Y = 80$$

Also,

The total number of students in the school in Jaipur = 180

The number of boys in the school in Jaipur = (A-55)

So, the number of girls in the school in Jaipur = (B+5)

Now,

$$(A-55) + (B+5) = 180$$

$$A + B = 230 \dots \dots \dots (III)$$

And,

$$A - B = 120 \dots \dots \dots (IV)$$

By solving both equations (III) and (IV), we get,

$$2B = 110$$

$$B = 55$$

Now, put the value of B in equation (III), we get,

$$A + B = 230$$

$$A + 55 = 230$$

$$A = 175$$

Delhi

The total number of students = 320

The number of boys = (X+50) = (150+50) = 200

The number of girls = (320-200) = 120

Mumbai

The total number of students = 240

The number of boys = $2Y = (2 \times 80) = 160$

The number of girls = (240-160) = 80

Chennai

The total number of students = 400

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The number of boys = $3Y = (3 \times 80) = 240$

The number of girls = $(400 - 240) = 160$

Kolkata

The total number of students = 300

The number of boys = $(X + 25) = (150 + 25) = 175$

The number of girls = $(300 - 175) = 125$

Jaipur

The total number of students = 180

The number of boys = $(A - 55) = (175 - 55) = 120$

The number of girls = $(180 - 120) = 60$

City	The total number of students (boys + girls)	The number of boys	The number of girls
Delhi	320	200	120
Mumbai	240	160	80
Chennai	400	240	160
Kolkata	300	175	125
Jaipur	180	120	60
Total	1440	895	545

Answer: D

Let the number of girls who newly joined the school in Mumbai be n .

So, the number of girls who left the school in Chennai = $2n$

Now, the number of girls in the school in Mumbai = $(80 + n)$

And, the number of girls in the school in Chennai = $(160 - 2n)$

So,

$$(80 + n) / (160 - 2n) = 5/6$$

$$480 + 6n = 800 - 10n$$

$$16n = 320$$

$$n = 20$$

Therefore, the number of girls who left the school in Chennai = $2n = (2 \times 20) = 40$.

Hence, the correct answer is option D.

3. Questions

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Delhi and Mumbai

The total number of students in the school in Delhi = 320

The number of boys in the school in Delhi = $(X+50)$

So, the number of girls in the school in Delhi = $\{320-(X+50)\} = (270-X)$

The total number of students in the school in Mumbai = 240

The number of boys in the school in Mumbai = $2Y$

So, the number of girls in the school in Mumbai = $(240-2Y)$

Chennai and Kolkata

The total number of students in the school in Chennai = 400

The number of boys in the school in Chennai = $3Y$

So, the number of girls in the school in Chennai = $(400-3Y)$

The total number of students in the school in Kolkata = 300

The number of boys in the school in Kolkata = $(X+25)$

So, the number of girls in the school in Kolkata = $\{300-(X+25)\} = (275-X)$

Now,

$$(270-X)/(240-2Y) = 150/100$$

$$(270-X)/(240-2Y) = 3/2$$

$$540 - 2X = 720 - 6Y$$

$$6Y - 2X = 180$$

$$3Y - X = 90 \dots\dots\dots(I)$$

And,

$$(400-3Y)/(275-X) = 32/25$$

$$10000 - 75Y = 8800 - 32X$$

$$75Y - 32X = 1200 \dots\dots\dots(II)$$

By solving both equations (I) and (II), we get,

$$225Y - 75X = 6750$$

$$225Y - 96X = 3600$$

$$21X = 3150$$

$$X = 150$$

Now, put the value of X in equation (I), we get,

$$3Y - X = 90$$

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$$3Y - 150 = 90$$

$$3Y = 240$$

$$Y = 80$$

Also,

The total number of students in the school in Jaipur = 180

The number of boys in the school in Jaipur = (A-55)

So, the number of girls in the school in Jaipur = (B+5)

Now,

$$(A-55) + (B+5) = 180$$

$$A + B = 230 \dots \dots \dots (III)$$

And,

$$A - B = 120 \dots \dots \dots (IV)$$

By solving both equations (III) and (IV), we get,

$$2B = 110$$

$$B = 55$$

Now, put the value of B in equation (III), we get,

$$A + B = 230$$

$$A + 55 = 230$$

$$A = 175$$

Delhi

The total number of students = 320

The number of boys = (X+50) = (150+50) = 200

The number of girls = (320-200) = 120

Mumbai

The total number of students = 240

The number of boys = 2Y = (2*80) = 160

The number of girls = (240-160) = 80

Chennai

The total number of students = 400

The number of boys = 3Y = (3*80) = 240

The number of girls = (400-240) = 160

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Kolkata

The total number of students = 300

The number of boys = $(X+25) = (150+25) = 175$

The number of girls = $(300-175) = 125$

Jaipur

The total number of students = 180

The number of boys = $(A-55) = (175-55) = 120$

The number of girls = $(180-120) = 60$

City	The total number of students (boys + girls)	The number of boys	The number of girls
Delhi	320	200	120
Mumbai	240	160	80
Chennai	400	240	160
Kolkata	300	175	125
Jaipur	180	120	60
Total	1440	895	545

Answer: A

Pune

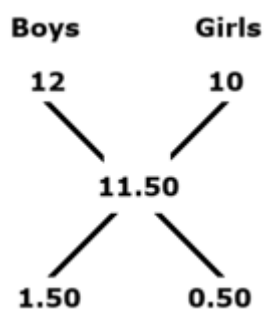
The total number of students = $(A+25) = (175+25) = 200$ (**A = 175, from common explanation**)

So, the average age of all students (boys + girls) = $(2300/200) = 11.50$ years

The average age of all boys = 12 years

The average age of all girls = 10 years

Now, using allegation method, we get,



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So, the ratio of the number of boys and girls in the school in Pune = $1.50:0.50 = 3:1$.

Therefore, the required difference = $2/4$ of $200 = 1/2$ of $200 = 100$.

Hence, the correct answer is option A.

4. Questions

Delhi and Mumbai

The total number of students in the school in Delhi = 320

The number of boys in the school in Delhi = $(X+50)$

So, the number of girls in the school in Delhi = $\{320-(X+50)\} = (270-X)$

The total number of students in the school in Mumbai = 240

The number of boys in the school in Mumbai = $2Y$

So, the number of girls in the school in Mumbai = $(240-2Y)$

Chennai and Kolkata

The total number of students in the school in Chennai = 400

The number of boys in the school in Chennai = $3Y$

So, the number of girls in the school in Chennai = $(400-3Y)$

The total number of students in the school in Kolkata = 300

The number of boys in the school in Kolkata = $(X+25)$

So, the number of girls in the school in Kolkata = $\{300-(X+25)\} = (275-X)$

Now,

$$(270-X)/(240-2Y) = 150/100$$

$$(270-X)/(240-2Y) = 3/2$$

$$540 - 2X = 720 - 6Y$$

$$6Y - 2X = 180$$

$$3Y - X = 90 \dots\dots\dots(I)$$

And,

$$(400-3Y)/(275-X) = 32/25$$

$$10000 - 75Y = 8800 - 32X$$

$$75Y - 32X = 1200 \dots\dots\dots(II)$$

By solving both equations (I) and (II), we get,

$$225Y - 75X = 6750$$

$$225Y - 96X = 3600$$

$$21X = 3150$$

$$X = 150$$

Now, put the value of X in equation (I), we get,

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$$3Y - X = 90$$

$$3Y - 150 = 90$$

$$3Y = 240$$

$$Y = 80$$

Also,

The total number of students in the school in Jaipur = 180

The number of boys in the school in Jaipur = (A-55)

So, the number of girls in the school in Jaipur = (B+5)

Now,

$$(A-55) + (B+5) = 180$$

$$A + B = 230 \dots \dots \dots (III)$$

And,

$$A - B = 120 \dots \dots \dots (IV)$$

By solving both equations (III) and (IV), we get,

$$2B = 110$$

$$B = 55$$

Now, put the value of B in equation (III), we get,

$$A + B = 230$$

$$A + 55 = 230$$

$$A = 175$$

Delhi

The total number of students = 320

The number of boys = (X+50) = (150+50) = 200

The number of girls = (320-200) = 120

Mumbai

The total number of students = 240

The number of boys = 2Y = (2*80) = 160

The number of girls = (240-160) = 80

Chennai

The total number of students = 400

The number of boys = 3Y = (3*80) = 240

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The number of girls = $(400-240) = 160$

Kolkata

The total number of students = 300

The number of boys = $(X+25) = (150+25) = 175$

The number of girls = $(300-175) = 125$

Jaipur

The total number of students = 180

The number of boys = $(A-55) = (175-55) = 120$

The number of girls = $(180-120) = 60$

City	The total number of students (boys + girls)	The number of boys	The number of girls
Delhi	320	200	120
Mumbai	240	160	80
Chennai	400	240	160
Kolkata	300	175	125
Jaipur	180	120	60
Total	1440	895	545

Answer: C

Kolkata

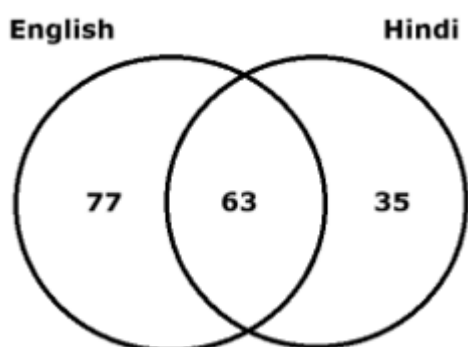
The total number of boys = 175

The number of boys who speaks both English and Hindi = 63

So, the number of boys who speaks only English = $11/16$ of $(175-63) = 11/16$ of 112 = 77

Therefore, the total number of boys who speaks English = $(77+63) = 140$.

Hence, the correct answer is option C.



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5. Questions

Delhi and Mumbai

The total number of students in the school in Delhi = 320

The number of boys in the school in Delhi = $(X+50)$

So, the number of girls in the school in Delhi = $\{320-(X+50)\} = (270-X)$

The total number of students in the school in Mumbai = 240

The number of boys in the school in Mumbai = $2Y$

So, the number of girls in the school in Mumbai = $(240-2Y)$

Chennai and Kolkata

The total number of students in the school in Chennai = 400

The number of boys in the school in Chennai = $3Y$

So, the number of girls in the school in Chennai = $(400-3Y)$

The total number of students in the school in Kolkata = 300

The number of boys in the school in Kolkata = $(X+25)$

So, the number of girls in the school in Kolkata = $\{300-(X+25)\} = (275-X)$

Now,

$$(270-X)/(240-2Y) = 150/100$$

$$(270-X)/(240-2Y) = 3/2$$

$$540 - 2X = 720 - 6Y$$

$$6Y - 2X = 180$$

$$3Y - X = 90 \dots\dots\dots(I)$$

And,

$$(400-3Y)/(275-X) = 32/25$$

$$10000 - 75Y = 8800 - 32X$$

$$75Y - 32X = 1200 \dots\dots\dots(II)$$

By solving both equations (I) and (II), we get,

$$225Y - 75X = 6750$$

$$225Y - 96X = 3600$$

$$21X = 3150$$

$$X = 150$$

Now, put the value of X in equation (I), we get,

$$3Y - X = 90$$

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$$3Y - 150 = 90$$

$$3Y = 240$$

$$Y = 80$$

Also,

The total number of students in the school in Jaipur = 180

The number of boys in the school in Jaipur = (A-55)

So, the number of girls in the school in Jaipur = (B+5)

Now,

$$(A-55) + (B+5) = 180$$

$$A + B = 230 \dots\dots\dots(III)$$

And,

$$A - B = 120 \dots\dots\dots(IV)$$

By solving both equations (III) and (IV), we get,

$$2B = 110$$

$$B = 55$$

Now, put the value of B in equation (III), we get,

$$A + B = 230$$

$$A + 55 = 230$$

$$A = 175$$

Delhi

The total number of students = 320

The number of boys = (X+50) = (150+50) = 200

The number of girls = (320-200) = 120

Mumbai

The total number of students = 240

The number of boys = 2Y = (2*80) = 160

The number of girls = (240-160) = 80

Chennai

The total number of students = 400

The number of boys = 3Y = (3*80) = 240

The number of girls = (400-240) = 160

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Kolkata

The total number of students = 300

The number of boys = $(X+25) = (150+25) = 175$

The number of girls = $(300-175) = 125$

Jaipur

The total number of students = 180

The number of boys = $(A-55) = (175-55) = 120$

The number of girls = $(180-120) = 60$

City	The total number of students (boys + girls)	The number of boys	The number of girls
Delhi	320	200	120
Mumbai	240	160	80
Chennai	400	240	160
Kolkata	300	175	125
Jaipur	180	120	60
Total	1440	895	545

Answer: E

Jaipur

The ratio of the number of boys and girls = $120:60 = 2:1$

So, the ratio of the number of male and female teachers in the school = $2:1$

The total number of teachers = $1/3$ of $180 = 60$

Therefore, the number of female teachers in the school in Jaipur = $1/3$ of $60 = 20$.

Hence, the correct answer is option E.

6. Questions

2015

@QuizByMickey

Let the initial investment made by P, Q and R be Rs. $3n$, Rs. $2n$ and Rs. $4n$ respectively.

First 5 months

$P = (3n \times 5) = \text{Rs. } 15n$

$Q = (2n \times 5) = \text{Rs. } 10n$

$R = (4n \times 5) = \text{Rs. } 20n$

Middle 4 months

$$P = \{(3n-200)*4\} = \text{Rs. } (12n-800)$$

$$Q = \{(2n+600)*4\} = \text{Rs. } (8n+2400)$$

$$R = \{(3/4 \text{ of } 4n)*4\} = \text{Rs. } 12n$$

Last 3 months

$$P = [\{150\% \text{ of } (3n-200)\}*3] = \{(4.50n-300)*3\} = \text{Rs. } (13.50n-900)$$

$$Q = \{(2n+600-300)*3\} = \{(2n+300)*3\} = \text{Rs. } (6n+900)$$

$$R = \{(3n+800)*3\} = \text{Rs. } (9n+2400)$$

The total investment made by

$$P = \{15n+(12n-800)+(13.50n-900)\} = \text{Rs. } (40.50n-1700)$$

$$Q = \{10n+(8n+2400)+(6n+900)\} = \text{Rs. } (24n+3300)$$

$$R = \{20n+12n+(9n+2400)\} = \text{Rs. } (41n+2400)$$

Now,

$$(41n+2400)/(24n+3300) = 188/129$$

$$(41n+2400)/(8n+1100) = 188/43$$

$$1763n + 103200 = 1504n + 206800$$

$$259n = 103600$$

$$n = 400$$

Therefore, the profit ratio of

$$P = (40.50n-1700) = (40.50*400-1700) = (16200-1700) = 14500$$

$$Q = (24n+3300) = (24*400+3300) = (9600+3300) = 12900$$

$$R = (41n+2400) = (41*400+2400) = (16400+2400) = 18800$$

So, the ratio of the profit share received by P, Q and R = 14500:12900:18800 = 145:129:188.

2016

Let the initial investment made by all three persons be T months.

First T months

$$P = (900*T) = \text{Rs. } 900T$$

$$Q = (700*T) = \text{Rs. } 700T$$

$$R = (1200*T) = \text{Rs. } 1200T$$

Last (12-T) months

$$P = \{(2/3 \text{ of } 900)*(12-T)\} = \{600*(12-T)\} = \text{Rs. } (7200-600T)$$

$$Q = \{(700+300)*(12-T)\} = \{1000*(12-T)\} = \text{Rs. } (12000-1000T)$$

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$$R = \{(75\% \text{ of } 1200) * (12 - T)\} = \{900 * (12 - T)\} = \text{Rs. } (10800 - 900T)$$

The total investment made by

$$P = \{900T + (7200 - 600T)\} = \text{Rs. } (7200 + 300T)$$

$$Q = \{700T + (12000 - 1000T)\} = \text{Rs. } (12000 - 300T)$$

$$R = \{1200T + (10800 - 900T)\} = \text{Rs. } (10800 + 300T)$$

Now,

$$(7200 + 300T) = (12000 - 300T)$$

$$600T = 4800$$

$$T = 8$$

Therefore, the profit ratio of

$$P = (7200 + 300T) = (7200 + 300 \times 8) = (7200 + 2400) = 9600$$

$$Q = (12000 - 300T) = (12000 - 2400) = 9600$$

$$R = (10800 + 300T) = (10800 + 2400) = 13200$$

So, the ratio of the profit share received by P, Q and R = 9600:9600:13200 = 8:8:11.

Answer: C

Now, according to the question,

2016

Let the total profit earned by all three persons be Rs. 100X.

So, the amount of profit donated to the charity = 10% of 100X = Rs. 10X

And, the profit share received by R = $\frac{11}{27}$ of $(100X - 10X) = \frac{11}{27}$ of 90X = Rs. $(\frac{110}{3})X$

Now,

$$(\frac{110}{3})X - 10X = 64000$$

$$(\frac{80}{3})X = 64000$$

$$X = 2400$$

Therefore, the total profit received by all three persons together in 2016 = 100X = (100×2400) = Rs. 240000.

Hence, the correct answer is option C.

7. Questions

2015

Let the initial investment made by P, Q and R be Rs. 3n, Rs. 2n and Rs. 4n respectively.

First 5 months

$$P = (3n \times 5) = \text{Rs. } 15n$$

$$Q = (2n \times 5) = \text{Rs. } 10n$$

$$R = (4n \times 5) = \text{Rs. } 20n$$

Middle 4 months

$$P = \{(3n-200) \times 4\} = \text{Rs. } (12n-800)$$

$$Q = \{(2n+600) \times 4\} = \text{Rs. } (8n+2400)$$

$$R = \{(3/4 \text{ of } 4n) \times 4\} = \text{Rs. } 12n$$

Last 3 months

$$P = [\{150\% \text{ of } (3n-200)\} \times 3] = \{(4.50n-300) \times 3\} = \text{Rs. } (13.50n-900)$$

$$Q = \{(2n+600-300) \times 3\} = \{(2n+300) \times 3\} = \text{Rs. } (6n+900)$$

$$R = \{(3n+800) \times 3\} = \text{Rs. } (9n+2400)$$

The total investment made by

$$P = \{15n + (12n-800) + (13.50n-900)\} = \text{Rs. } (40.50n-1700)$$

$$Q = \{10n + (8n+2400) + (6n+900)\} = \text{Rs. } (24n+3300)$$

$$R = \{20n + 12n + (9n+2400)\} = \text{Rs. } (41n+2400)$$

Now,

$$(41n+2400)/(24n+3300) = 188/129$$

$$(41n+2400)/(8n+1100) = 188/43$$

$$1763n + 103200 = 1504n + 206800$$

$$259n = 103600$$

$$n = 400$$

Therefore, the profit ratio of

$$P = (40.50n-1700) = (40.50 \times 400 - 1700) = (16200 - 1700) = 14500$$

$$Q = (24n+3300) = (24 \times 400 + 3300) = (9600 + 3300) = 12900$$

$$R = (41n+2400) = (41 \times 400 + 2400) = (16400 + 2400) = 18800$$

So, the ratio of the profit share received by P, Q and R = 14500:12900:18800 = 145:129:188.

2016

Let the initial investment made by all three persons be T months.

First T months

$$P = (900 \times T) = \text{Rs. } 900T$$

$$Q = (700 \times T) = \text{Rs. } 700T$$

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$$R = (1200 * T) = \text{Rs. } 1200T$$

Last (12-T) months

$$P = \{(2/3 \text{ of } 900) * (12-T)\} = \{600 * (12-T)\} = \text{Rs. } (7200-600T)$$

$$Q = \{(700+300) * (12-T)\} = \{1000 * (12-T)\} = \text{Rs. } (12000-1000T)$$

$$R = \{(75\% \text{ of } 1200) * (12-T)\} = \{900 * (12-T)\} = \text{Rs. } (10800-900T)$$

The total investment made by

$$P = \{900T + (7200-600T)\} = \text{Rs. } (7200+300T)$$

$$Q = \{700T + (12000-1000T)\} = \text{Rs. } (12000-300T)$$

$$R = \{1200T + (10800-900T)\} = \text{Rs. } (10800+300T)$$

Now,

$$(7200+300T) = (12000-300T)$$

$$600T = 4800$$

$$T = 8$$

Therefore, the profit ratio of

$$P = (7200+300T) = (7200+300 \times 8) = (7200+2400) = 9600$$

$$Q = (12000-300T) = (12000-2400) = 9600$$

$$R = (10800+300T) = (10800+2400) = 13200$$

So, the ratio of the profit share received by P, Q and R = 9600:9600:13200 = 8:8:11.

Answer: D

I: The initial investment made by A in 2015 = $3n = (3 \times 400) = \text{Rs. } 1200$.

So, I is correct.

II: The difference between the initial investment made by B and C in 2015 = $(4n-2n) = 2n = (2 \times 400) = \text{Rs. } 800$.

So, II is correct.

Hence, the correct answer is option D.

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8. Questions

2015

Let the initial investment made by P, Q and R be Rs. $3n$, Rs. $2n$ and Rs. $4n$ respectively.

First 5 months

$$P = (3n \times 5) = \text{Rs. } 15n$$

$$Q = (2n \times 5) = \text{Rs. } 10n$$

$$R = (4n \times 5) = \text{Rs. } 20n$$

Middle 4 months

$$P = \{(3n-200) \times 4\} = \text{Rs. } (12n-800)$$

$$Q = \{(2n+600) \times 4\} = \text{Rs. } (8n+2400)$$

$$R = \{(3/4 \text{ of } 4n) \times 4\} = \text{Rs. } 12n$$

Last 3 months

$$P = [\{150\% \text{ of } (3n-200)\} \times 3] = \{(4.50n-300) \times 3\} = \text{Rs. } (13.50n-900)$$

$$Q = \{(2n+600-300) \times 3\} = \{(2n+300) \times 3\} = \text{Rs. } (6n+900)$$

$$R = \{(3n+800) \times 3\} = \text{Rs. } (9n+2400)$$

The total investment made by

$$P = \{15n + (12n-800) + (13.50n-900)\} = \text{Rs. } (40.50n-1700)$$

$$Q = \{10n + (8n+2400) + (6n+900)\} = \text{Rs. } (24n+3300)$$

$$R = \{20n + 12n + (9n+2400)\} = \text{Rs. } (41n+2400)$$

Now,

$$(41n+2400)/(24n+3300) = 188/129$$

$$(41n+2400)/(8n+1100) = 188/43$$

$$1763n + 103200 = 1504n + 206800$$

$$259n = 103600$$

$$n = 400$$

Therefore, the profit ratio of

$$P = (40.50n-1700) = (40.50 \times 400 - 1700) = (16200 - 1700) = 14500$$

$$Q = (24n+3300) = (24 \times 400 + 3300) = (9600 + 3300) = 12900$$

$$R = (41n+2400) = (41 \times 400 + 2400) = (16400 + 2400) = 18800$$

So, the ratio of the profit share received by P, Q and R = 14500:12900:18800 = 145:129:188.

2016

Let the initial investment made by all three persons be T months.

First T months

$$P = (900 \times T) = \text{Rs. } 900T$$

$$Q = (700 \times T) = \text{Rs. } 700T$$

$$R = (1200 \times T) = \text{Rs. } 1200T$$

Last (12-T) months

$$P = \{(2/3 \text{ of } 900) * (12-T)\} = \{600 * (12-T)\} = \text{Rs. } (7200-600T)$$

$$Q = \{(700+300) * (12-T)\} = \{1000 * (12-T)\} = \text{Rs. } (12000-1000T)$$

$$R = \{(75\% \text{ of } 1200) * (12-T)\} = \{900 * (12-T)\} = \text{Rs. } (10800-900T)$$

The total investment made by

$$P = \{900T + (7200-600T)\} = \text{Rs. } (7200+300T)$$

$$Q = \{700T + (12000-1000T)\} = \text{Rs. } (12000-300T)$$

$$R = \{1200T + (10800-900T)\} = \text{Rs. } (10800+300T)$$

Now,

$$(7200+300T) = (12000-300T)$$

$$600T = 4800$$

$$T = 8$$

Therefore, the profit ratio of

$$P = (7200+300T) = (7200+300 \times 8) = (7200+2400) = 9600$$

$$Q = (12000-300T) = (12000-2400) = 9600$$

$$R = (10800+300T) = (10800+2400) = 13200$$

So, the ratio of the profit share received by P, Q and R = 9600:9600:13200 = 8:8:11.

Answer: C

2016

The investment made by S in first 9 months = $(4Y \times 9) = \text{Rs. } 36Y$

And, the investment made by S in last 3 months = $\{(3/4 \text{ of } 4Y) \times 3\} = (3Y \times 3) = \text{Rs. } 9Y$

So, the total investment made by S = $(36Y + 9Y) = \text{Rs. } 45Y$

Now,

$$9600/45Y = 16/15$$

$$720Y = 144000$$

$$Y = 200$$

Therefore, the required value = 20% of $2Y = 20\% \text{ of } (2 \times 200) = 20\% \text{ of } 400 = 80$.

Hence, the correct answer is option C.

9. Questions

2015

Let the initial investment made by P, Q and R be Rs. 3n, Rs. 2n and Rs. 4n respectively.

First 5 months

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$$P = (3n \times 5) = \text{Rs. } 15n$$

$$Q = (2n \times 5) = \text{Rs. } 10n$$

$$R = (4n \times 5) = \text{Rs. } 20n$$

Middle 4 months

$$P = \{(3n-200) \times 4\} = \text{Rs. } (12n-800)$$

$$Q = \{(2n+600) \times 4\} = \text{Rs. } (8n+2400)$$

$$R = \{(3/4 \text{ of } 4n) \times 4\} = \text{Rs. } 12n$$

Last 3 months

$$P = [\{150\% \text{ of } (3n-200)\} \times 3] = \{(4.50n-300) \times 3\} = \text{Rs. } (13.50n-900)$$

$$Q = \{(2n+600-300) \times 3\} = \{(2n+300) \times 3\} = \text{Rs. } (6n+900)$$

$$R = \{(3n+800) \times 3\} = \text{Rs. } (9n+2400)$$

The total investment made by

$$P = \{15n + (12n-800) + (13.50n-900)\} = \text{Rs. } (40.50n-1700)$$

$$Q = \{10n + (8n+2400) + (6n+900)\} = \text{Rs. } (24n+3300)$$

$$R = \{20n + 12n + (9n+2400)\} = \text{Rs. } (41n+2400)$$

Now,

$$(41n+2400)/(24n+3300) = 188/129$$

$$(41n+2400)/(8n+1100) = 188/43$$

$$1763n + 103200 = 1504n + 206800$$

$$259n = 103600$$

$$n = 400$$

Therefore, the profit ratio of

$$P = (40.50n-1700) = (40.50 \times 400 - 1700) = (16200 - 1700) = 14500$$

$$Q = (24n+3300) = (24 \times 400 + 3300) = (9600 + 3300) = 12900$$

$$R = (41n+2400) = (41 \times 400 + 2400) = (16400 + 2400) = 18800$$

So, the ratio of the profit share received by P, Q and R = 14500:12900:18800 = 145:129:188.

2016

Let the initial investment made by all three persons be T months.

First T months

$$P = (900 \times T) = \text{Rs. } 900T$$

$$Q = (700 \times T) = \text{Rs. } 700T$$

@QuizByMickey

$$R = (1200 \times T) = \text{Rs. } 1200T$$

Last (12-T) months

$$P = \left\{ \left(\frac{2}{3} \text{ of } 900 \right) \times (12-T) \right\} = \{ 600 \times (12-T) \} = \text{Rs. } (7200-600T)$$

$$Q = \{ (700+300) \times (12-T) \} = \{ 1000 \times (12-T) \} = \text{Rs. } (12000-1000T)$$

$$R = \{ (75\% \text{ of } 1200) \times (12-T) \} = \{ 900 \times (12-T) \} = \text{Rs. } (10800-900T)$$

The total investment made by

$$P = \{ 900T + (7200-600T) \} = \text{Rs. } (7200+300T)$$

$$Q = \{ 700T + (12000-1000T) \} = \text{Rs. } (12000-300T)$$

$$R = \{ 1200T + (10800-900T) \} = \text{Rs. } (10800+300T)$$

Now,

$$(7200+300T) = (12000-300T)$$

$$600T = 4800$$

$$T = 8$$

Therefore, the profit ratio of

$$P = (7200+300T) = (7200+300 \times 8) = (7200+2400) = 9600$$

$$Q = (12000-300T) = (12000-2400) = 9600$$

$$R = (10800+300T) = (10800+2400) = 13200$$

So, the ratio of the profit share received by P, Q and R = 9600:9600:13200 = 8:8:11.

Answer: B

2017

The total investment made by P = $(500 \times 12) = \text{Rs. } 6000$

And, the total investment made by Q = $\{ 1000 \times (12-Z) \} = \text{Rs. } (12000-1000Z)$

Now,

$$6000 / (12000-1000Z) = 6/7$$

$$72000 - 6000Z = 42000$$

$$6000Z = 30000$$

$$Z = 5$$

Therefore, the number of months after which Q joined the business = $Z = 5$ months.

Hence, the correct answer is option B.

10. Questions

2015

@QuizByMickey

Let the initial investment made by P, Q and R be Rs. $3n$, Rs. $2n$ and Rs. $4n$ respectively.

First 5 months

$$P = (3n \times 5) = \text{Rs. } 15n$$

$$Q = (2n \times 5) = \text{Rs. } 10n$$

$$R = (4n \times 5) = \text{Rs. } 20n$$

Middle 4 months

$$P = \{(3n - 200) \times 4\} = \text{Rs. } (12n - 800)$$

$$Q = \{(2n + 600) \times 4\} = \text{Rs. } (8n + 2400)$$

$$R = \{(3/4 \text{ of } 4n) \times 4\} = \text{Rs. } 12n$$

Last 3 months

$$P = [\{150\% \text{ of } (3n - 200)\} \times 3] = \{(4.50n - 300) \times 3\} = \text{Rs. } (13.50n - 900)$$

$$Q = \{(2n + 600 - 300) \times 3\} = \{(2n + 300) \times 3\} = \text{Rs. } (6n + 900)$$

$$R = \{(3n + 800) \times 3\} = \text{Rs. } (9n + 2400)$$

The total investment made by

$$P = \{15n + (12n - 800) + (13.50n - 900)\} = \text{Rs. } (40.50n - 1700)$$

$$Q = \{10n + (8n + 2400) + (6n + 900)\} = \text{Rs. } (24n + 3300)$$

$$R = \{20n + 12n + (9n + 2400)\} = \text{Rs. } (41n + 2400)$$

Now,

$$(41n + 2400) / (24n + 3300) = 188/129$$

$$(41n + 2400) / (8n + 1100) = 188/43$$

$$1763n + 103200 = 1504n + 206800$$

$$259n = 103600$$

$$n = 400$$

Therefore, the profit ratio of

$$P = (40.50n - 1700) = (40.50 \times 400 - 1700) = (16200 - 1700) = 14500$$

$$Q = (24n + 3300) = (24 \times 400 + 3300) = (9600 + 3300) = 12900$$

$$R = (41n + 2400) = (41 \times 400 + 2400) = (16400 + 2400) = 18800$$

So, the ratio of the profit share received by P, Q and R = $14500:12900:18800 = 145:129:188$.

2016

Let the initial investment made by all three persons be T months.

First T months

$$P = (900 \times T) = \text{Rs. } 900T$$

$$Q = (700 \times T) = \text{Rs. } 700T$$

$$R = (1200 \times T) = \text{Rs. } 1200T$$

Last (12-T) months

$$P = \left\{ \left(\frac{2}{3} \text{ of } 900 \right) \times (12-T) \right\} = \{ 600 \times (12-T) \} = \text{Rs. } (7200-600T)$$

$$Q = \{ (700+300) \times (12-T) \} = \{ 1000 \times (12-T) \} = \text{Rs. } (12000-1000T)$$

$$R = \{ (75\% \text{ of } 1200) \times (12-T) \} = \{ 900 \times (12-T) \} = \text{Rs. } (10800-900T)$$

The total investment made by

$$P = \{ 900T + (7200-600T) \} = \text{Rs. } (7200+300T)$$

$$Q = \{ 700T + (12000-1000T) \} = \text{Rs. } (12000-300T)$$

$$R = \{ 1200T + (10800-900T) \} = \text{Rs. } (10800+300T)$$

Now,

$$(7200+300T) = (12000-300T)$$

$$600T = 4800$$

$$T = 8$$

Therefore, the profit ratio of

$$P = (7200+300T) = (7200+300 \times 8) = (7200+2400) = 9600$$

$$Q = (12000-300T) = (12000-2400) = 9600$$

$$R = (10800+300T) = (10800+2400) = 13200$$

So, the ratio of the profit share received by P, Q and R = 9600:9600:13200 = 8:8:11.

Answer: A

2015

The ratio of the profit share of P, Q and R = 145:129:188

Therefore, the profit received by Q = $\left[\frac{43000}{(188-145)} \times 129 \right] = \left\{ \frac{(43000/43) \times 129}{1} \right\} = \text{Rs. } 129000$.

Hence, the correct answer is option A.

11. Questions

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = 2P% of 2000 = 40P

The number of voters who did not cast their votes = (2000-40P)

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P - 1400)$

Now,

$$(40P - 1400)/(2000 - 40P) = 1/2$$

$$80P - 2800 = 2000 - 40P$$

$$120P = 4800$$

$$P = 40$$

Village B

The total number of voters registered = $100Q$

The number of voters who casted their votes = 90% of $100Q = 90Q$

The number of voters who did not cast their votes = $(100Q - 90Q) = 10Q$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q - 960)$

Now,

$$(90Q - 960) = 10Q$$

$$80Q = 960$$

$$Q = 12$$

Also,

Let the number of valid votes casted by females in village A be k .

So, the number of valid votes casted by males in village B = k

The number of valid votes casted by males in village A = $(1400 - k)$

The number of valid votes casted by females in village B = $(960 - k)$

Now,

$$(1400 - k)/(960 - k) = 100/45$$

$$(1400 - k)/(960 - k) = 20/9$$

$$12600 - 9k = 19200 - 20k$$

$$11k = 6600$$

$$k = 600$$

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $40P = (40 \times 40) = 1600$

The number of voters who did not cast their votes = $(2000 - 40P) = (2000 - 1600) = 400$

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The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P - 1400) = (1600 - 1400) = 200$

The number of valid votes casted by females = $k = 600$

The number of valid votes casted by males = $(1400 - 600) = 800$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 4:3$

$M:3 = 4:3$

$M = 4$

Village B

The total number of voters registered = $100Q = (100 \times 12) = 1200$

The number of voters who casted their votes = $90Q = (90 \times 12) = 1080$

The number of voters who did not cast their votes = $10Q = (10 \times 12) = 120$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q - 960) = (1080 - 960) = 120$

The number of valid votes casted by males = $k = 600$

The number of valid votes casted by females = $(960 - 600) = 360$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 600:360 = 5:3$

$N:3 = 5:3$

$N = 5$

Village	The total number of voters registered	The number of voters who casted their votes	The number of voters who did not cast their votes	The number of valid votes casted by the voters	The number of invalid votes casted by the voters	The number of valid votes casted by males	The number of invalid votes casted by females
A	2000	1600	400	1400	200	800	600
B	1200	1080	120	960	120	600	360
Total	3200	2680	520	2360	320	1400	960

Answer: D

Now, according to the question,

Village A

The total number of votes casted (valid + invalid) = 1600

The total number of votes casted (valid + invalid) by males = $19/32$ of $1600 = 950$

The total number of votes casted (valid + invalid) by females = $13/32$ of $1600 = 650$

The number of valid votes casted by males = 800

The number of valid votes casted by females = 600

So, the number of invalid votes casted by males = $(950-800) = 150$

And, the number of invalid votes casted by females = $(650-600) = 50$

Therefore, the required difference = $(150-50) = 100$.

Hence, the correct answer is option D.

12. Questions

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $2P\%$ of $2000 = 40P$

The number of voters who did not cast their votes = $(2000-40P)$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P-1400)$

Now,

$$(40P-1400)/(2000-40P) = 1/2$$

$$80P - 2800 = 2000 - 40P$$

$$120P = 4800$$

$$P = 40$$

Village B

The total number of voters registered = $100Q$

The number of voters who casted their votes = 90% of $100Q = 90Q$

The number of voters who did not cast their votes = $(100Q-90Q) = 10Q$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q-960)$

Now,

$$(90Q-960) = 10Q$$

$$80Q = 960$$

$$Q = 12$$

Also,

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Let the number of valid votes casted by females in village A be k .

So, the number of valid votes casted by males in village B = k

The number of valid votes casted by males in village A = $(1400-k)$

The number of valid votes casted by females in village B = $(960-k)$

Now,

$$(1400-k)/(960-k) = 100/45$$

$$(1400-k)/(960-k) = 20/9$$

$$12600 - 9k = 19200 - 20k$$

$$11k = 6600$$

$$k = 600$$

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $40P = (40 \times 40) = 1600$

The number of voters who did not cast their votes = $(2000-40P) = (2000-1600) = 400$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P-1400) = (1600-1400) = 200$

The number of valid votes casted by females = $k = 600$

The number of valid votes casted by males = $(1400-600) = 800$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 4:3$

$$M:3 = 4:3$$

$$M = 4$$

Village B

The total number of voters registered = $100Q = (100 \times 12) = 1200$

The number of voters who casted their votes = $90Q = (90 \times 12) = 1080$

The number of voters who did not cast their votes = $10Q = (10 \times 12) = 120$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q-960) = (1080-960) = 120$

The number of valid votes casted by males = $k = 600$

The number of valid votes casted by females = $(960-600) = 360$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 600:360 = 5:3$

$$N:3 = 5:3$$

N = 5

Village	The total number of voters registered	The number of voters who casted their votes	The number of voters who did not cast their votes	The number of valid votes casted by the voters	The number of invalid votes casted by the voters	The number of valid votes casted by males	The number of invalid votes casted by females
A	2000	1600	400	1400	200	800	600
B	1200	1080	120	960	120	600	360
Total	3200	2680	520	2360	320	1400	960

Answer: A

Village B

The total number of valid votes casted = 960

The total number of invalid votes casted = 120

The ratio of the number of valid and invalid votes casted to party X = 5:1. Let it be 5t and t respectively.

So, the number of valid votes casted to party Y = (960-5t)

And, the number of invalid votes casted to party Y = (120-t)

Now,

$$(960-5t) - (120-t) = 440$$

$$520 - 5t = 120 - t$$

$$4t = 400$$

$$t = 100$$

So, the total number of votes casted (valid + invalid) to party X = (5t+t) = 6t = (6*100) = 600

And, the total number of votes casted (valid + invalid) to party Y = {(960-5t)+(120-t)} = (1080-6t) = (1080-600) = 480

Therefore, the required ratio = 600:480 = 5:4.

Hence, the correct answer is option A.

13. Questions

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = 2P% of 2000 = 40P

The number of voters who did not cast their votes = $(2000-40P)$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P-1400)$

Now,

$$(40P-1400)/(2000-40P) = 1/2$$

$$80P - 2800 = 2000 - 40P$$

$$120P = 4800$$

$$P = 40$$

Village B

The total number of voters registered = $100Q$

The number of voters who casted their votes = 90% of $100Q = 90Q$

The number of voters who did not cast their votes = $(100Q-90Q) = 10Q$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q-960)$

Now,

$$(90Q-960) = 10Q$$

$$80Q = 960$$

$$Q = 12$$

Also,

Let the number of valid votes casted by females in village A be k .

So, the number of valid votes casted by males in village B = k

The number of valid votes casted by males in village A = $(1400-k)$

The number of valid votes casted by females in village B = $(960-k)$

Now,

$$(1400-k)/(960-k) = 100/45$$

$$(1400-k)/(960-k) = 20/9$$

$$12600 - 9k = 19200 - 20k$$

$$11k = 6600$$

$$k = 600$$

Village A

The total number of voters registered = 2000

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The number of voters who casted their votes = $40P = (40 \times 40) = 1600$

The number of voters who did not cast their votes = $(2000 - 40P) = (2000 - 1600) = 400$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P - 1400) = (1600 - 1400) = 200$

The number of valid votes casted by females = $k = 600$

The number of valid votes casted by males = $(1400 - 600) = 800$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 4:3$

$M:3 = 4:3$

$M = 4$

Village B

The total number of voters registered = $100Q = (100 \times 12) = 1200$

The number of voters who casted their votes = $90Q = (90 \times 12) = 1080$

The number of voters who did not cast their votes = $10Q = (10 \times 12) = 120$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q - 960) = (1080 - 960) = 120$

The number of valid votes casted by males = $k = 600$

The number of valid votes casted by females = $(960 - 600) = 360$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 600:360 = 5:3$

$N:3 = 5:3$

$N = 5$

Village	The total number of voters registered	The number of voters who casted their votes	The number of voters who did not cast their votes	The number of valid votes casted by the voters	The number of invalid votes casted by the voters	The number of valid votes casted by males	The number of invalid votes casted by females
A	2000	1600	400	1400	200	800	600
B	1200	1080	120	960	120	600	360
Total	3200	2680	520	2360	320	1400	960

Answer: B

Village C

The total number of voters who casted their votes = $150N = (150 \times 5) = 750$

The number of voters who casted invalid votes = $1/5$ of $750 = 150$

The number of voters who casted valid votes = $(750 - 150) = 600$

The number of voters who did not cast their votes = $(150 + 100) = 250$

So, the total number of voters registered in village C = $(750 + 250) = 1000$

And, the total number of voters registered in village B = 1200

Therefore, the required percentage = $\{(1000/1200) \times 100\} \% = 83.33\%$.

Hence, the correct answer is option B.

14. Questions

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $2P\%$ of $2000 = 40P$

The number of voters who did not cast their votes = $(2000 - 40P)$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P - 1400)$

Now,

$$(40P - 1400) / (2000 - 40P) = 1/2$$

$$80P - 2800 = 2000 - 40P$$

$$120P = 4800$$

$$P = 40$$

Village B

The total number of voters registered = $100Q$

The number of voters who casted their votes = 90% of $100Q = 90Q$

The number of voters who did not cast their votes = $(100Q - 90Q) = 10Q$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q - 960)$

Now,

$$(90Q - 960) = 10Q$$

$$80Q = 960$$

$$Q = 12$$

Also,

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Let the number of valid votes casted by females in village A be k .

So, the number of valid votes casted by males in village B = k

The number of valid votes casted by males in village A = $(1400-k)$

The number of valid votes casted by females in village B = $(960-k)$

Now,

$$(1400-k)/(960-k) = 100/45$$

$$(1400-k)/(960-k) = 20/9$$

$$12600 - 9k = 19200 - 20k$$

$$11k = 6600$$

$$k = 600$$

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $40P = (40 \times 40) = 1600$

The number of voters who did not cast their votes = $(2000-40P) = (2000-1600) = 400$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P-1400) = (1600-1400) = 200$

The number of valid votes casted by females = $k = 600$

The number of valid votes casted by males = $(1400-600) = 800$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 4:3$

$$M:3 = 4:3$$

$$M = 4$$

Village B

The total number of voters registered = $100Q = (100 \times 12) = 1200$

The number of voters who casted their votes = $90Q = (90 \times 12) = 1080$

The number of voters who did not cast their votes = $10Q = (10 \times 12) = 120$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q-960) = (1080-960) = 120$

The number of valid votes casted by males = $k = 600$

The number of valid votes casted by females = $(960-600) = 360$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 600:360 = 5:3$

$$N:3 = 5:3$$

N = 5

Village	The total number of voters registered	The number of voters who casted their votes	The number of voters who did not cast their votes	The number of valid votes casted by the voters	The number of invalid votes casted by the voters	The number of valid votes casted by males	The number of invalid votes casted by females
A	2000	1600	400	1400	200	800	600
B	1200	1080	120	960	120	600	360
Total	3200	2680	520	2360	320	1400	960

Answer: C

I: $(P/N) = (40/5) = 8$

So, I is definitely true.

II: $(M+5) = (4+5) = 9$

$Q = 12$

So, II is definitely not true.

III: $(P-M) = (40-4) = 36$

$6^2 = 36$

So, III is definitely true.

Therefore, both I and III are definitely true.

Hence, the correct answer is option C.

15. Questions

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $2P\%$ of 2000 = $40P$

The number of voters who did not cast their votes = $(2000-40P)$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P-1400)$

Now,

$(40P-1400)/(2000-40P) = 1/2$

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$$80P - 2800 = 2000 - 40P$$

$$120P = 4800$$

$$P = 40$$

Village B

The total number of voters registered = $100Q$

The number of voters who casted their votes = 90% of $100Q = 90Q$

The number of voters who did not cast their votes = $(100Q - 90Q) = 10Q$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q - 960)$

Now,

$$(90Q - 960) = 10Q$$

$$80Q = 960$$

$$Q = 12$$

Also,

Let the number of valid votes casted by females in village A be k .

So, the number of valid votes casted by males in village B = k

The number of valid votes casted by males in village A = $(1400 - k)$

The number of valid votes casted by females in village B = $(960 - k)$

Now,

$$(1400 - k)/(960 - k) = 100/45$$

$$(1400 - k)/(960 - k) = 20/9$$

$$12600 - 9k = 19200 - 20k$$

$$11k = 6600$$

$$k = 600$$

Village A

The total number of voters registered = 2000

The number of voters who casted their votes = $40P = (40 \times 40) = 1600$

The number of voters who did not cast their votes = $(2000 - 40P) = (2000 - 1600) = 400$

The number of valid votes casted by the voters = 1400

The number of invalid votes casted = $(40P - 1400) = (1600 - 1400) = 200$

The number of valid votes casted by females = $k = 600$

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The number of valid votes casted by males = $(1400-600) = 800$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 4:3$

$M:3 = 4:3$

$M = 4$

Village B

The total number of voters registered = $100Q = (100 \times 12) = 1200$

The number of voters who casted their votes = $90Q = (90 \times 12) = 1080$

The number of voters who did not cast their votes = $10Q = (10 \times 12) = 120$

The number of valid votes casted by the voters = 960

The number of invalid votes casted = $(90Q-960) = (1080-960) = 120$

The number of valid votes casted by males = $k = 600$

The number of valid votes casted by females = $(960-600) = 360$

So, the ratio of the number of valid votes casted by males and females = $800:600 = 600:360 = 5:3$

$N:3 = 5:3$

$N = 5$

Village	The total number of voters registered	The number of voters who casted their votes	The number of voters who did not cast their votes	The number of valid votes casted by the voters	The number of invalid votes casted by the voters	The number of valid votes casted by males	The number of invalid votes casted by females
A	2000	1600	400	1400	200	800	600
B	1200	1080	120	960	120	600	360
Total	3200	2680	520	2360	320	1400	960

Answer: D

Village A

The total number of valid votes casted = 1400

The number of votes casted to losing candidate = $1/8$ of 1600 = 200 (**Note: $12.50\% = 1/8$**)

The number of votes casted to winning candidate = $(1400-200) = 1200$

Therefore, the number of votes by which winning candidate won the election in village A = $(1200-200) = 1000$.

Hence, the correct answer is option D.

16. Questions

Answer: E

Equation I: $A^3 - 28A^2 + 187A = 0$

$$A (A^2 - 28A + 187) = 0$$

$$A (A^2 - 17A - 11A + 187) = 0$$

$$A [A(A-17) - 11(A-17)] = 0$$

$$A (A-11) (A-17) = 0$$

$$A = 0 \text{ or } +11 \text{ or } +17$$

$$\text{So, } A = 0 \text{ or } 11 \text{ or } 17$$

Equation II: $B^3 - 32B^2 + 192B = 0$ (divide by B)

$$B (B^2 - 32B + 192) = 0$$

$$B (B^2 - 24B - 8B + 192) = 0$$

$$B [B(B-24) - 8(B-24)] = 0$$

$$B (B-8) (B-24) = 0$$

$$B = 0 \text{ or } +8 \text{ or } +24$$

$$\text{So, } B = 0 \text{ or } 8 \text{ or } 24$$

Value of 'A'	Relation	Value of 'B'
0	=	0
0	<	8
0	<	24
11	>	0
11	>	8
11	<	24
17	>	0
17	>	8
17	<	24

Therefore, relation between A and B cannot be determined.

Hence, the correct answer is option E.

17. Questions

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Answer: B

The possible values of $M = 3, 5$ or 7

Case I

Let $M = 3$

Now,

$$(M-1)^3 = N$$

$$(3-1)^3 = 2^3 = 8$$

$N = 8$ (possible)

Case II

Let $M = 5$

Now,

$$(M-1)^3 = N$$

$$(5-1)^3 = 4^3 = 64$$

$N = 64$ (not possible)

Case III

Let $M = 7$

Now,

$$(M-1)^3 = N$$

$$(7-1)^3 = 6^3 = 216$$

$N = 216$ (not possible)

So, case I is satisfied.

Therefore,

$$M = 3$$

$$N = 8$$

Equation I: $P^2 + MP - 108 = 0$

$$P^2 + 3P - 108 = 0$$

$$P^2 + 12P - 9P - 108 = 0$$

$$P(P+12) - 9(P+12) = 0$$

$$(P-9)(P+12) = 0$$

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$$P = +9 \text{ or } -12$$

$$\text{So, } P = 9 \text{ or } -12$$

$$\text{Smaller root} = -12$$

$$\text{Equation II: } Q^2 - NQ - 240 = 0$$

$$Q^2 - 8Q - 240 = 0$$

$$Q^2 + 12Q - 20Q - 240 = 0$$

$$Q(Q+12) - 20(Q+12) = 0$$

$$(Q-20)(Q+12) = 0$$

$$Q = +20 \text{ or } -12$$

$$\text{So, } Q = 20 \text{ or } -12$$

$$\text{Smaller root} = -12$$

$$\text{Therefore, the required difference} = \{-12 - (-12)\} = (-12 + 12) = 0.$$

Hence, the correct answer is option B.

18. Questions

Answer: A

$$\text{Equation: } K^2 - 17K + 70 = 0$$

$$K^2 - 10K - 7K + 70 = 0$$

$$K(K-10) - 7(K-10) = 0$$

$$(K-7)(K-10) = 0$$

$$K = +7 \text{ or } +10$$

$$\text{So, the smaller root} = 7$$

Option a)

$$L^2 - 11L + 28 = 0$$

$$L^2 - 7L - 4L + 28 = 0$$

$$L(L-7) - 4(L-7) = 0$$

$$(L-4)(L-7) = 0$$

$$L = +4 \text{ or } +7$$

Here, 7 is greater root.

So, option a) does satisfy.

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Option b)

$$L^2 - 18L + 77 = 0$$

$$L^2 - 11L - 7L + 77 = 0$$

$$L(L-11) - 7(L-11) = 0$$

$$(L-7)(L-11) = 0$$

$$L = +7 \text{ or } +11$$

Here, 7 is smaller root.

So, option b) does not satisfy.

Option c)

$$L^2 - 27L + 140 = 0$$

$$L^2 - 20L - 7L + 140 = 0$$

$$L(L-20) - 7(L-20) = 0$$

$$(L-7)(L-20) = 0$$

$$L = +7 \text{ or } +20$$

Here, 7 is smaller root.

So, option c) does not satisfy.

Hence, the correct answer is option A.

19. Questions

Answer: D

Let the greater root be B.

The smaller root = (A-40)

The sum of the roots = A

Now,

$$B + (A-40) = A$$

$$A - B = A - 40$$

$$B = 40$$

Therefore, the greater root of the equation = B = 40.

Hence, the correct answer is option D.

20. Questions

Answer: A

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The smaller root = 5

The possible values of greater root are = 14, 41, 23, 32 or 50.

$$(1+4) = 5$$

$$(4+1) = 5$$

$$(2+3) = 5$$

$$(3+2) = 5$$

$$(5+0) = 5$$

So, the greater root = 23 (*more than 20 but less than 30*)

The sum of the roots = $M = (23+5) = 28$

And, the product of the roots = $N = (23*5) = 115$

Therefore, the value of $[N/(M-5)] = [115/(28-5)] = (115/23) = 5$.

Hence, the correct answer is option A.

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